Game Theory: Introduction Class

Indraprastha Institute of Information Technology, Delhi

April 7th, 2026

Why Game Theory and What to Expect?

- Is it useful for you?
- Who uses Game Theory?
  - Economists.
  - Machine Learning, AI (Shapley values, multi-agent systems).
  - Political Scientists.
  - Sports Strategists (Google "Penalty Kicks and Game Theory in the 2008 Champions League")
  - Most Decision Scientists (What is that? Examples?).

What is Game Theory?

- Consider two cases:
  1. You are ordering food on Zomato, and have 500 rupees at your disposal. You are trying to pick the best set of food items to order.
  2. You are taking a penalty kick in a football match. There is a goalkeeper in front of you.
- What is one fundamental difference between the above scenarios?

Let us define a game in general

- Who are the players?

$$i \in \{1, ..., I\} \text{ where } I \geq 2$$

- Strategies?

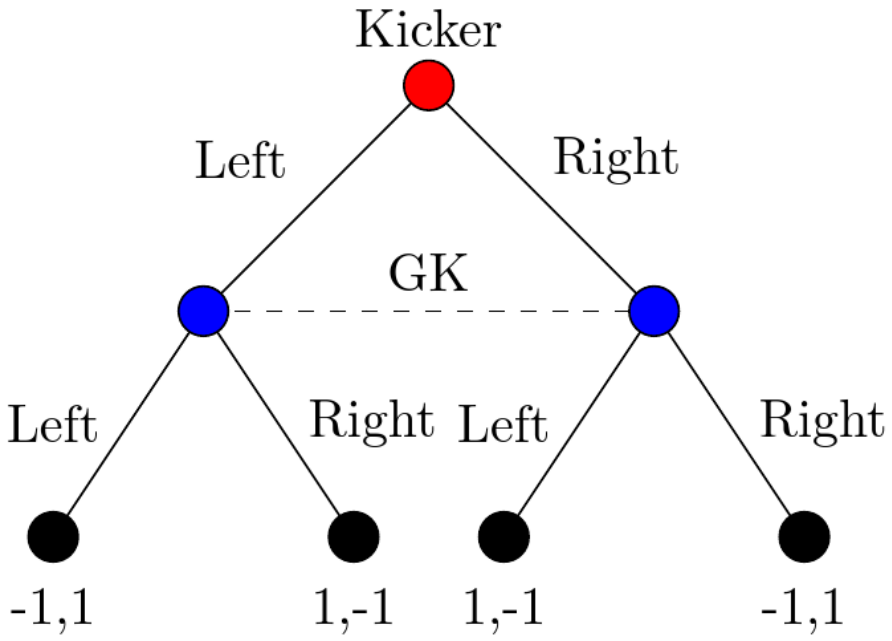
$$S \in \{S_1, ...S_I\}$$

- Payoffs?
  - What a player receives for a particular outcome (a good example is profit a firm receives in different scenarios).

Structure of the game

- Who are the players?
- What are the actions available to them?
- What is the time structure of the game?
- Payoff functions
- Common knowledge

Normal vs Extensive Form Game: Penalties



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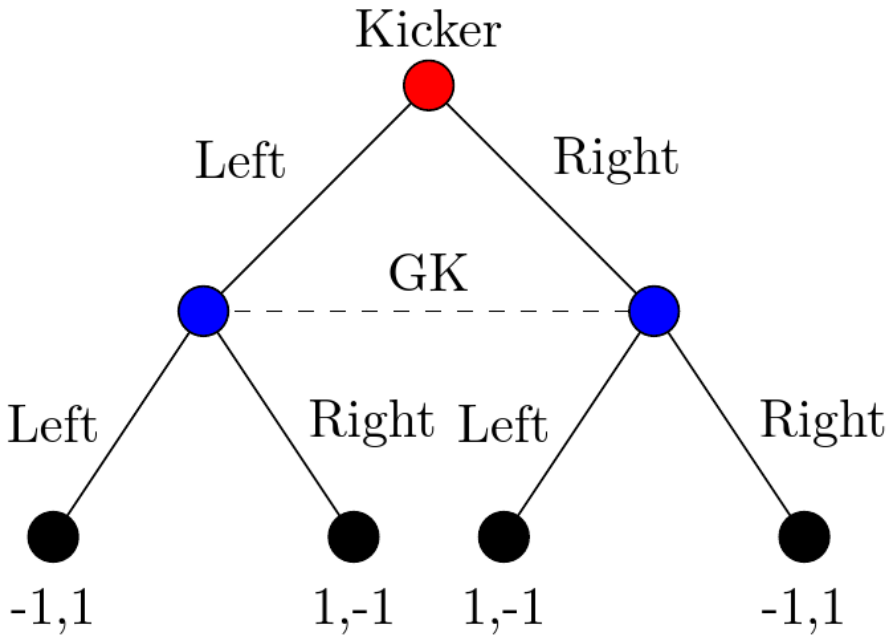

		GK	
		Left	Right
Kicker	Left	-1,1	1,-1
	Right	1,-1	-1,1

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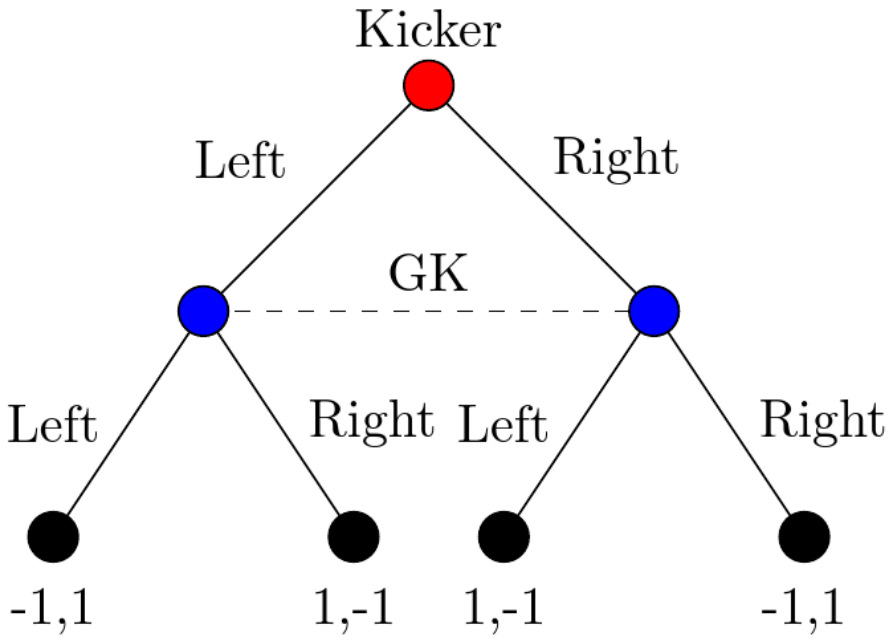
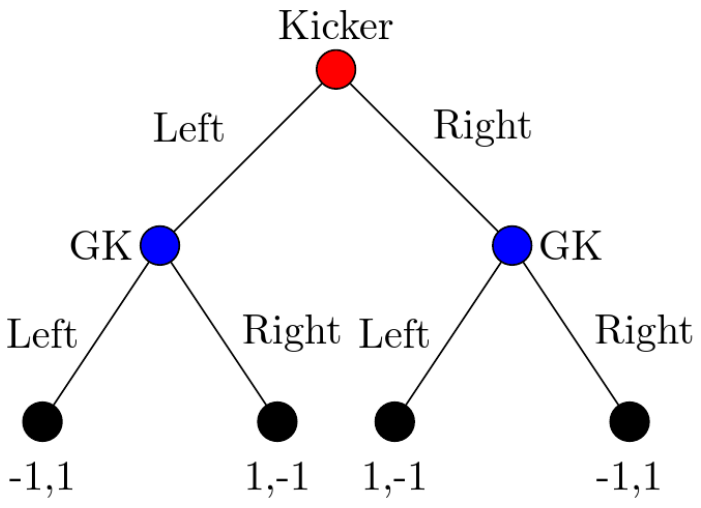


		GK	
		Left	Right
Kicker	Left	-1,1	1,-1
	Right	1,-1	-1,1

- $S_{\text{Kicker}} = \{\text{Left,Right}\}$, $S_{\text{GK}} = \{\text{Left,Right}\}$

Amateur Kicker vs World Class Goalkeeper

- This is a bit more difficult to write in Normal Form. Let us first look at the Extensive Form.



- What about the strategies?
- $S_{\text{Kicker}} = \{\text{Left,Right}\}$

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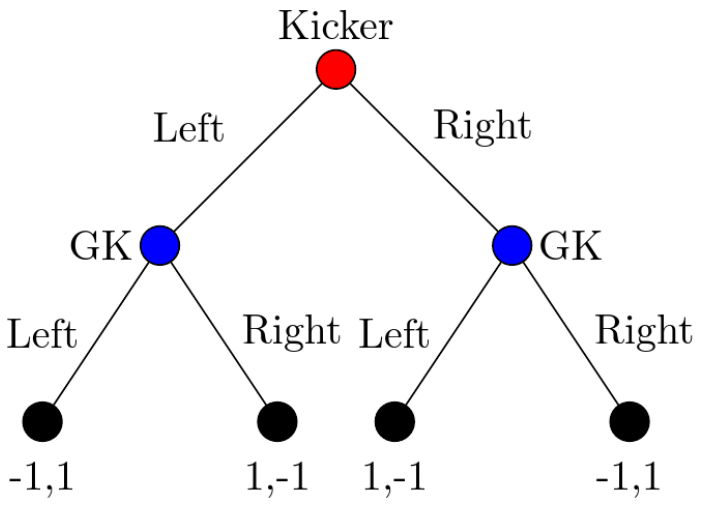
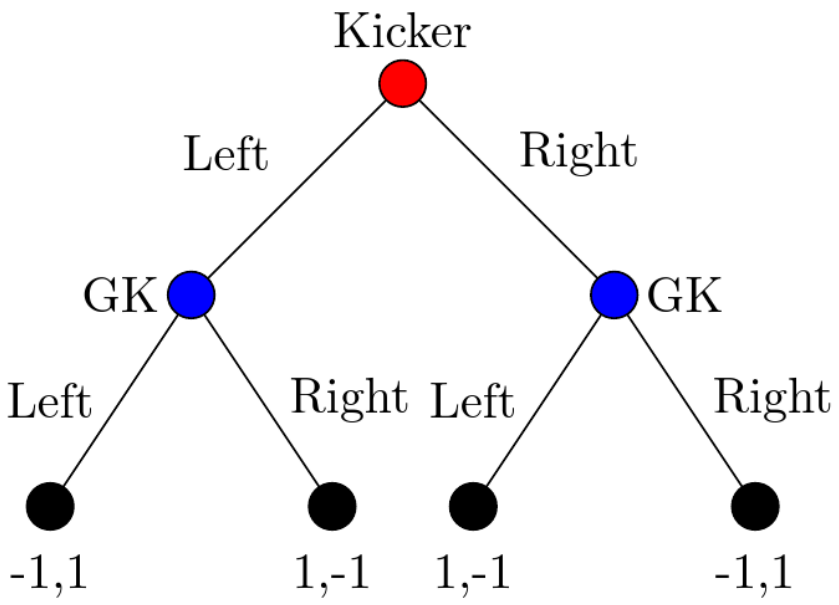


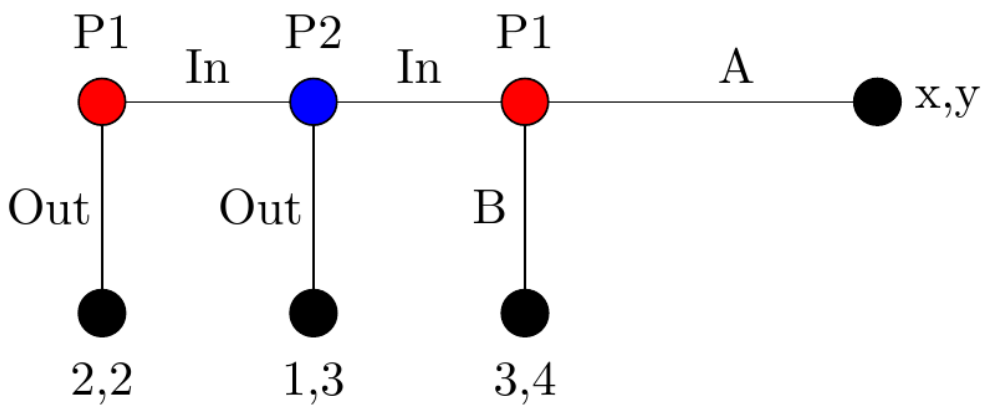
- What about the strategies?
- $S_{\text{Kicker}} = \{\text{Left,Right}\}$
- $S_{\text{GK}} = \{\text{LeftLeft,RightRight,LeftRight,RightLeft,RightRight}\}$

Extensive Form vs Normal Form



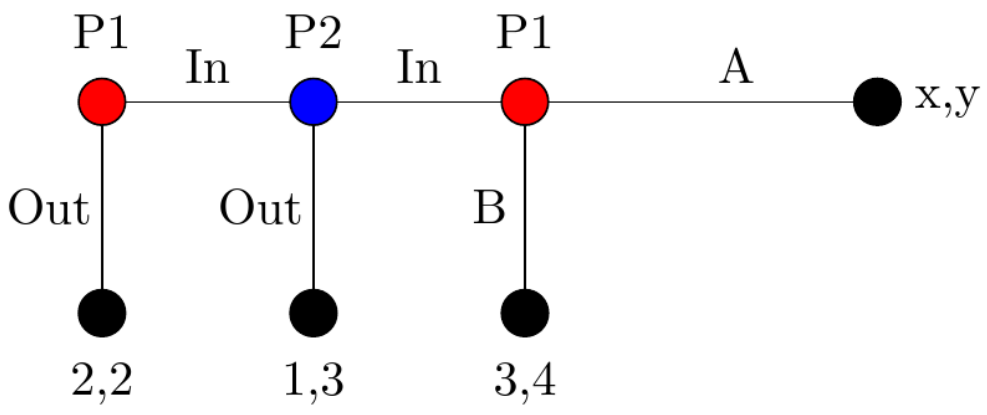
		GK			
		LeftLeft	LeftRight	RightLeft	RightRight
Kicker	Left	-1,1	-1,1	1,-1	1,-1
	Right	1,-1	-1,1	1,-1	-1,1

Let's try the Centipede Game



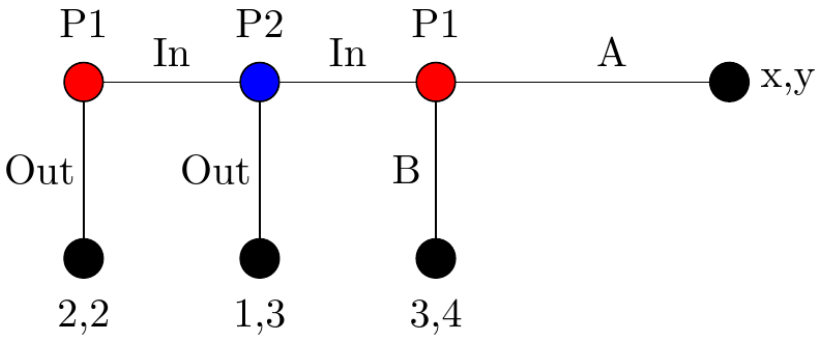
- Let's try to figure out the strategies available to each player. After that, the normal form is not too hard.

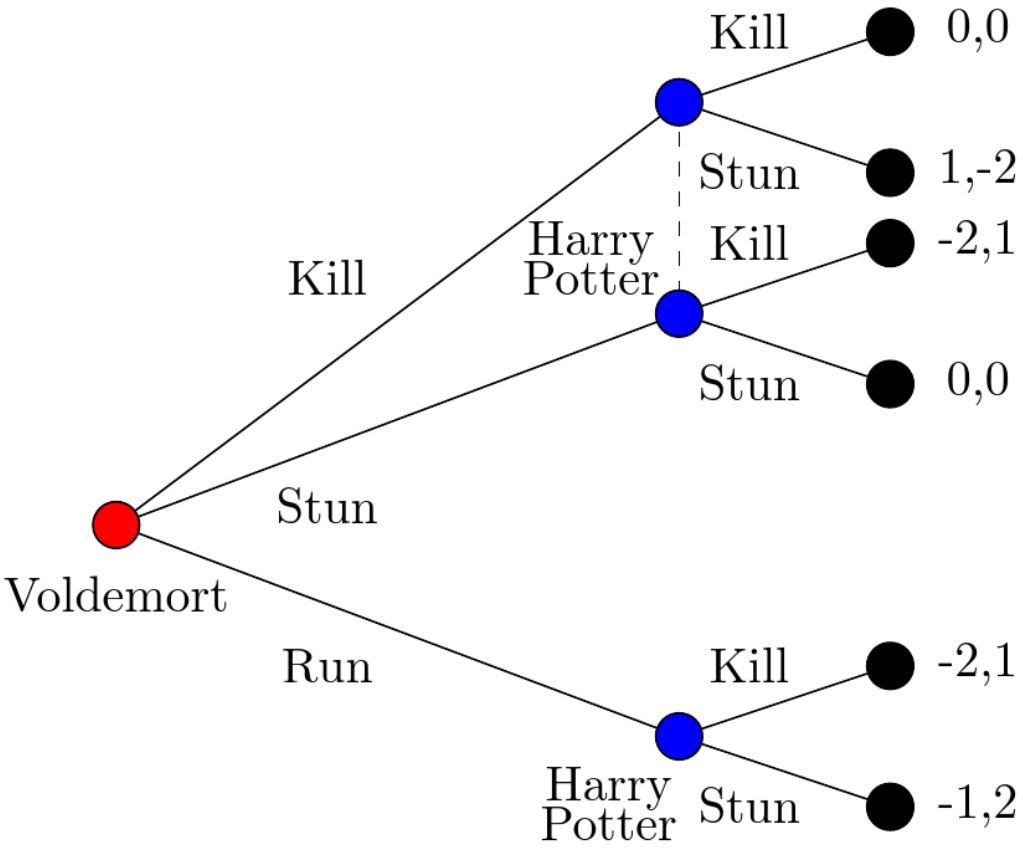
Let's try the Centipede Game



- Let's try to figure out the strategies available to each player. After that, the normal form is not too hard.
- $S_{\text{P1}} = \{\text{InA,InB,OutA,OutB}\}$, $S_{\text{P2}} = \{\text{In,Out}\}$.

Extensive Form vs Normal Form



		P2	
		In	Out
P1	InA	x,y	1,3
	InB	3,4	1,3
	OutA	2,2	2,2
	OutB	2,2	2,2

Harry Potter vs Voldemort: Try it!